UNIVERSITY OF LEEDS

COMP1860 Building our Digital World: Computer Systems & Architecture

S1.W3.L2: Boolean Logic & Logic Gates

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Overview

- Boolean Logic
- Logic Circuits

Boolean Algebra

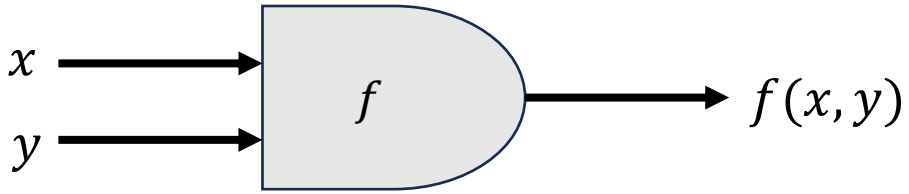
- Mathematical logic
  - Each Boolean variable has 2 possible values: True and False
- Named after George Boole
  - English mathematician
  - Proposed basic principles of the algebra in 1854

Number of possible values

- 1 variable: 2 possible values, 0 or 1
- 2 variables: 4 possible combinations – 00, 01, 10, 11
- 3 variables: 8 possible combinations – 000, 001, 010, 011, 100, 101, 110, 111
- *n* variables: 2^n possible combinations

Boolean functions

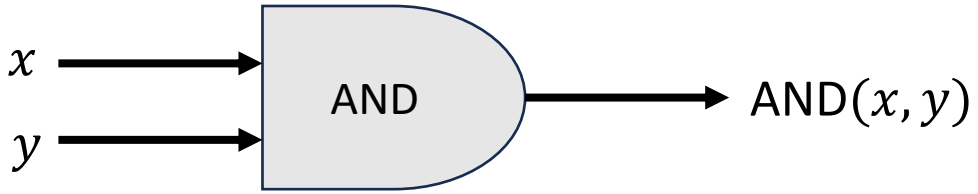
x	y	f
0	0	0
0	1	0
1	0	0
1	1	1



$$f(x, y) = \begin{cases} 1 \text{ when } x = 1 \text{ and } y = 1 \\ 0 \text{ otherwise} \end{cases}$$

Boolean functions

x	y	AND
0	0	0
0	1	0
1	0	0
1	1	1



$$\text{AND}(x, y) = \begin{cases} 1 \text{ when } x = 1 \text{ and } y = 1 \\ 0 \text{ otherwise} \end{cases}$$

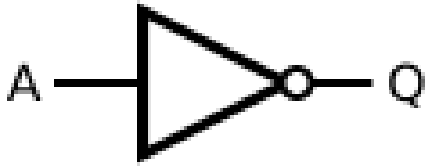
- Boolean function
  - A function that operates on Boolean variables, and returns a Boolean value
- Simple Boolean functions (like AND):
  - Sometimes called *operators（运算符）*
  - Their variables are called *operands（操作数）*
- $f(x, y)$ can also be written as $x\ f\ y$

Truth tables

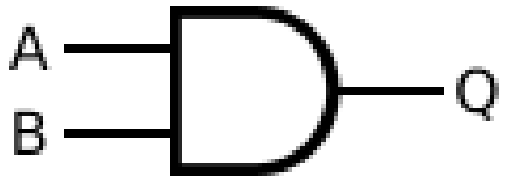
- The table from the previous slides is an example of a truth table
- It show all possible combinations of the inputs
- Output for each combination

x	y	AND
0	0	0
0	1	0
1	0	0
1	1	1

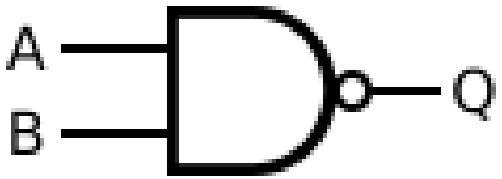
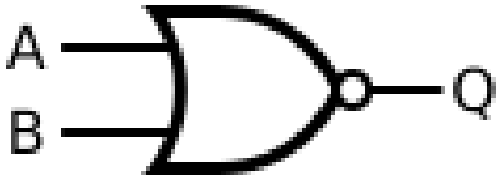
Boolean Operators

Operator	Boolean Algebra	Logic Gate Symbol	Truth Table
NOT	$\neg A$		A | ¬A 0 | 1 1 | 0

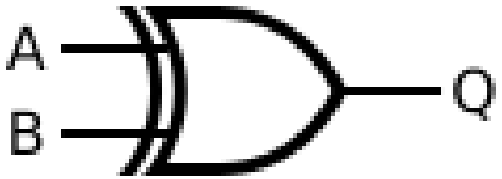
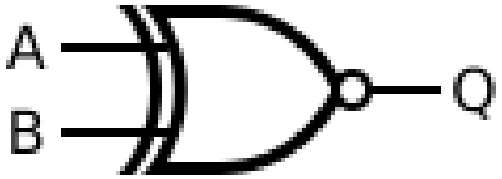
Boolean Operators

<table>
<tr><th>Operator</th><th>Boolean Algebra</th><th>Logic Gate Symbol</th><th>Truth Table</th></tr>
<tr><td>AND</td><td>$A \wedge B$</td><td> 
</td><td>
<table>
<tr><th>A</th><th>B</th><th>$A \wedge B$</th></tr>
<tr><td>0</td><td>0</td><td>0</td></tr>
<tr><td>0</td><td>1</td><td>0</td></tr>
<tr><td>1</td><td>0</td><td>0</td></tr>
<tr><td>1</td><td>1</td><td>1</td></tr>
</table>
</td></tr>
<tr><td>OR</td><td>$A \vee B$</td><td> 
</td><td>
<table>
<tr><th>A</th><th>B</th><th>$A \vee B$</th></tr>
<tr><td>0</td><td>0</td><td>0</td></tr>
<tr><td>0</td><td>1</td><td>1</td></tr>
<tr><td>1</td><td>0</td><td>1</td></tr>
<tr><td>1</td><td>1</td><td>1</td></tr>
</table>
</td></tr>
</table>

Boolean Operators

Operator	Boolean Algebra	Logic Gate Symbol	Truth Table
NAND （与非门）	$A \overline{\wedge} B$ or $A \uparrow B$ 真值表不用记直接推 ¬(A ∧ B)		A $\overline{\wedge}$ B 0 0 1 0 1 1 1 0 1 1 1 0
NOR （或非门）	$A \overline{\vee} B$ or $A \downarrow B$ 真值表不用记直接推 ¬(A ∨ B)		A $\overline{\vee}$ B 0 0 1 0 1 0 1 0 0 1 1 0

Boolean Operators

Operator	Boolean Algebra	Logic Gate Symbol	Truth Table
XOR (异或门)	$A \oplus B$ 相异为1		A B $A \oplus B$ 0 0 0 0 1 1 1 0 1 1 1 0
XNOR (同或门)	$A \odot B$ 相同为1		A B $A \odot B$ 0 0 1 0 1 0 1 0 0 1 1 1

Boolean Logic

- 通过组合operators， 可以得到具有更多variables的更复杂的expressions
  - E.g., $(A \wedge B) \vee C$ or $(\neg A \vee B) \wedge C$
- To evaluate:
  - Set the values of the variables
  - If ¬ then negate
  - Evaluate each section, ***clause***, (in parentheses) to true or false
  - Then evaluate the expression substituting in those values
- E.g., $(A \wedge B) \vee C$, A is 1, B is 0, C is 1
  - $A \wedge B$ evaluates as $1 \wedge 0$ which is 0.
  - Set $A \wedge B$ as 0 and evaluate $0 \vee 1$, which is 1.

Truth Tables for Expressions

- A truth table shows the evaluation of a Boolean formula for all possible input values
  - If n variables then 2^n rows
- E.g., truth table for $A \wedge \neg B$

A	B	A	$\wedge$	$\neg B$	$A \wedge \neg B$
0	0	0	0	1	0
0	1	0	0	0	0
1	0	1	1	1	1
1	1	1	0	0	0

Disjunctive Normal Form (DNF 1行内交外或)

表达式的形式写成如下形式

$$C_1 \vee C_2 \vee C_3 \vee \cdots \vee C_n$$

其中每个 C_i 内部形式如下

$$l_1 \wedge l_2 \wedge \cdots \wedge l_m$$

并且每一个 l_i is a variable or its negation, i.e. A or $\neg A$

For DNF, each term is an AND, terms are joined together by OR.

括号内用and连接，括号间用or连接 （内交外或）

Examples

The following are in DNF

- $(A \wedge \neg B \wedge \neg C) \vee (\neg D \wedge E \wedge F)$
- $(A \wedge B) \vee (C)$
- $(A \wedge B)$
- (A)

上式中每一个l_i is a variable or its negation

The following are not in DNF

- $\neg(A \vee B)$
  - $\neg(A \vee B)$不符合要求，多了¬
- $\neg(A \wedge B) \vee C$
  - $\neg(A \wedge B)$不符合要求，多了¬
- $A \vee \big(B \wedge (C \vee D)\big)$
  - $(C \vee D)$不符合要求，内交外或

Converting to DNF

- Use truth table

1. 挑选输出为1的行
2. 如果行中的输入为0，则取反相应的变量

Note: 这不会产生最小的DNF表达式!

Example

A	B	C	$(A \vee B) \wedge C$	
0	0	0	0	
0	0	1	0	
0	1	0	0	
0	1	1	1	1
1	0	0	0	
1	0	1	1	2
1	1	0	0	
1	1	1	1	3

1. 挑选输出为1的行，列为0要取反

- Row 1 → $\neg A \wedge B \wedge C$
- Row 2 → $A \wedge \neg B \wedge C$
- Row 3 → $A \wedge B \wedge C$

2. 把挑选中行用V 连接

- DNF

$$(\neg A \wedge B \wedge C) \vee (A \wedge \neg B \wedge C) \vee (A \wedge B \wedge C)$$

Exercise

- Using the truth tables you created earlier, convert the following to DNF:

1. $(A \wedge B) \vee C$
2. $(\neg A \vee B) \wedge C$

Conjunctive Normal Form (CNF 0行内或外交)

表达式的形式写成如下形式

$$C_1 \wedge C_2 \wedge C_3 \wedge \cdots \wedge C_n$$

其中每个 C_i 内部形式如下

$$l_1 \vee l_2 \vee \cdots \vee l_m$$

并且每一个 l_i is a variable or its negation, i.e. A or $\neg A$

For CNF, each term is an OR, terms are joined together by AND.

括号内用or连接，括号间用and连接 （内或外交）

Examples

The following are in CNF

- $(A \vee \neg B \vee \neg C) \wedge (\neg D \vee E \vee F)$
- $(A \vee B) \wedge (C)$
- $(A \vee B)$
- (A)

The following are not in CNF

- $\neg(A \wedge B)$
  - The AND is nested inside a NOT
- $\neg(A \vee B) \wedge C$
  - The OR is nested inside a NOT
- $A \wedge \big(B \vee (C \wedge D)\big)$
  - An AND is nested inside an OR

Converting to CNF

- Use truth table

1. 挑选输出为0的行
2. 如果行中的输入为1，则相应的变量取反
3. Note:这不会产生最小的CNF表达式!

Example

A	B	C	$(A \vee B) \wedge C$	
0	0	0	0	1
0	0	1	0	2
0	1	0	0	3
0	1	1	1	
1	0	0	0	4
1	0	1	1	
1	1	0	0	5
1	1	1	1	

挑选输出为0的行，列为1取反

- Row 1 → $A \vee B \vee C$
- Row 2 → $A \vee B \vee \neg C$
- Row 3 → $A \vee \neg B \vee C$
- Row 4 → $\neg A \vee B \vee C$
- Row 5→ $\neg A \vee \neg B \vee C$

挑选出的行and连接起来

- CNF

$(A \vee B \vee C) \wedge (A \vee B \vee \neg C)$
$\wedge (A \vee \neg B \vee C) \wedge (\neg A \vee B \vee C)$
$\wedge (\neg A \vee \neg B \vee C)$

Exercise

- Using the truth tables you created earlier, convert the following to CNF:

1. $(A \wedge B) \vee C$
2. $(\neg A \vee B) \wedge C$

DNF vs CNF

	连接规则	挑选结果行	输入取反规则	助记
• DNF	内交外或	结果为 1	为0取反	1 or 1
• CNF	内或外交	结果为 0	为1取反	0 and 0

DNF和CNF是等价的，即 ¬DNF = CNF。

DNF: $(\neg A \wedge B \wedge C) \vee (A \wedge \neg B \wedge C) = 1$

CNF: $\neg DNF = \neg 1$

$\neg(\neg A \wedge B \wedge C) \wedge \neg(A \wedge \neg B \wedge C) = 0$

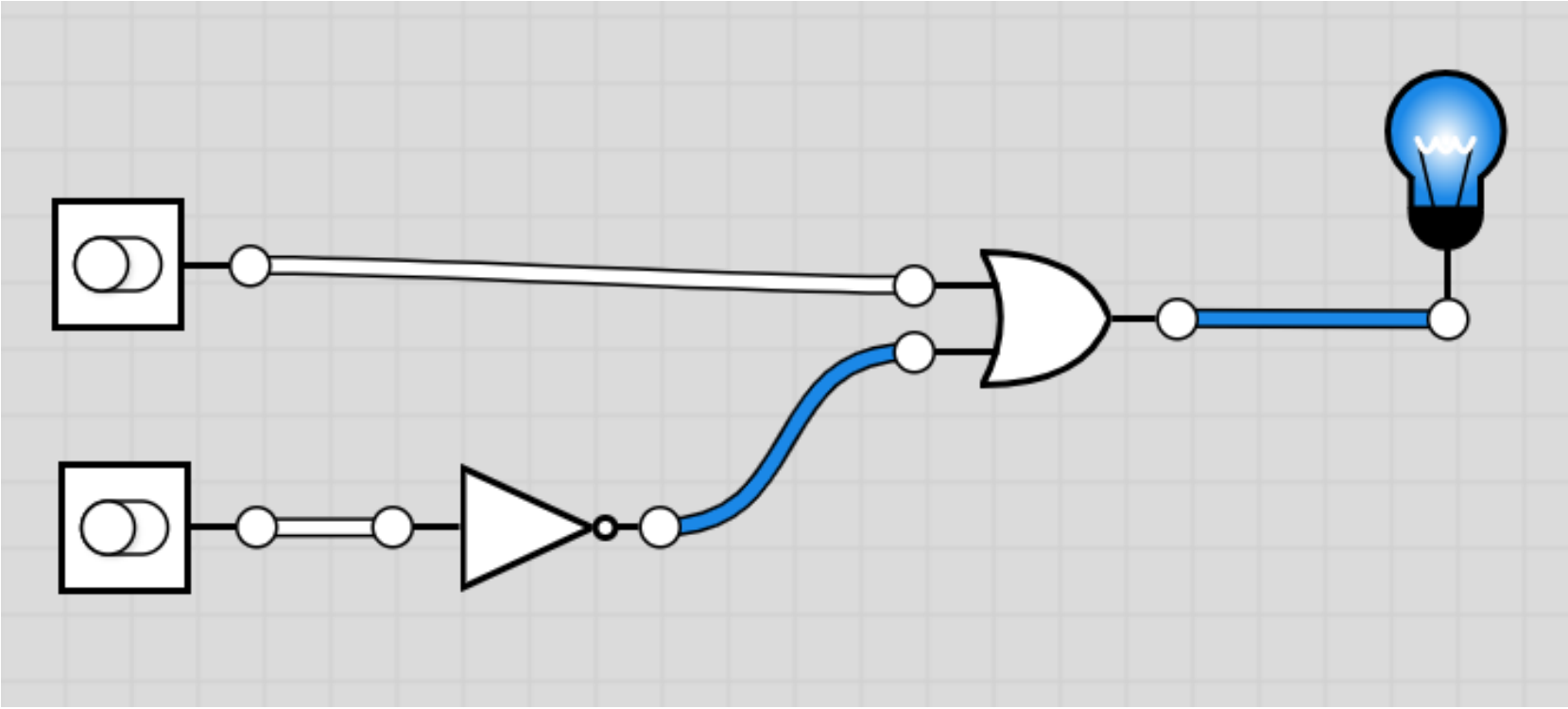
$(A \vee \neg B \vee \neg C) \wedge (\neg A \vee B \vee \neg C) = 0$

Logic Gates（逻辑门）

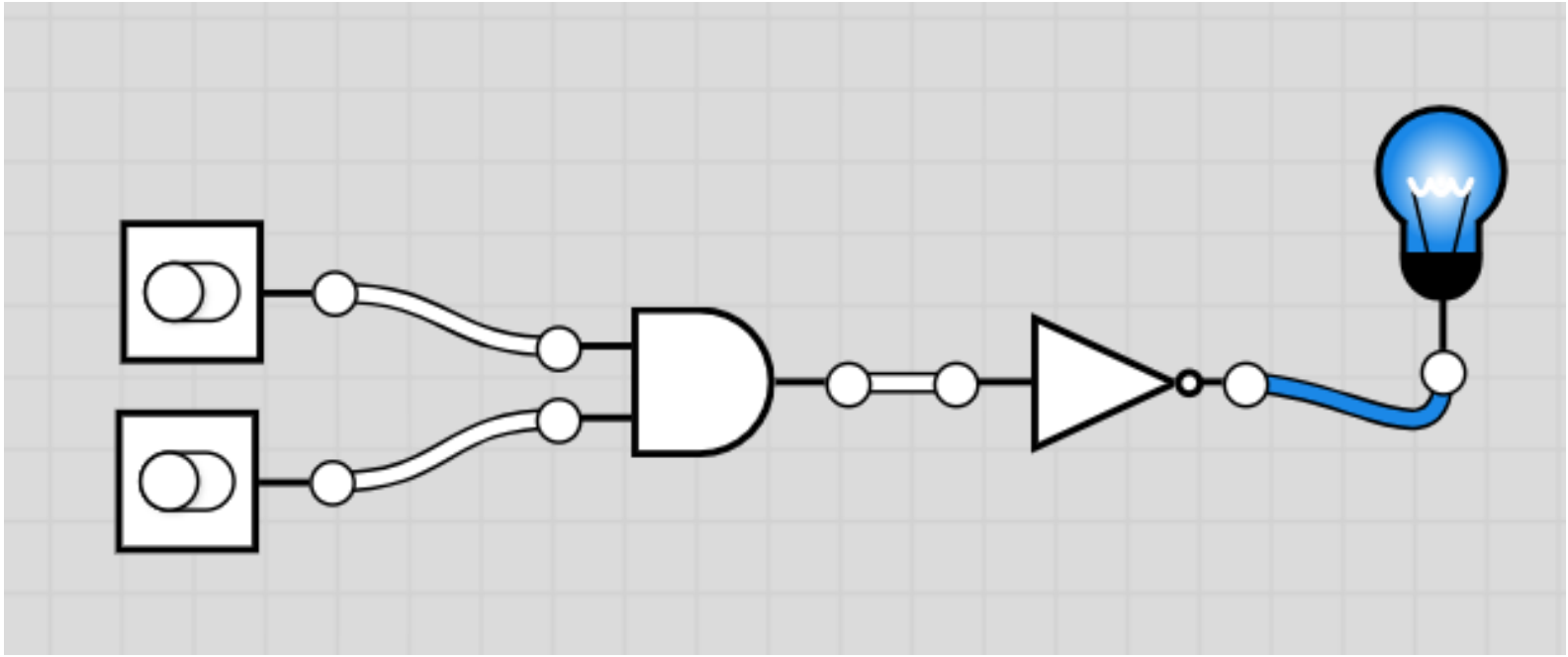
Logic Circuits（逻辑电路）

- Logic circuits are composed of logic gates wired together
- Remember each gate corresponds to a logical operator

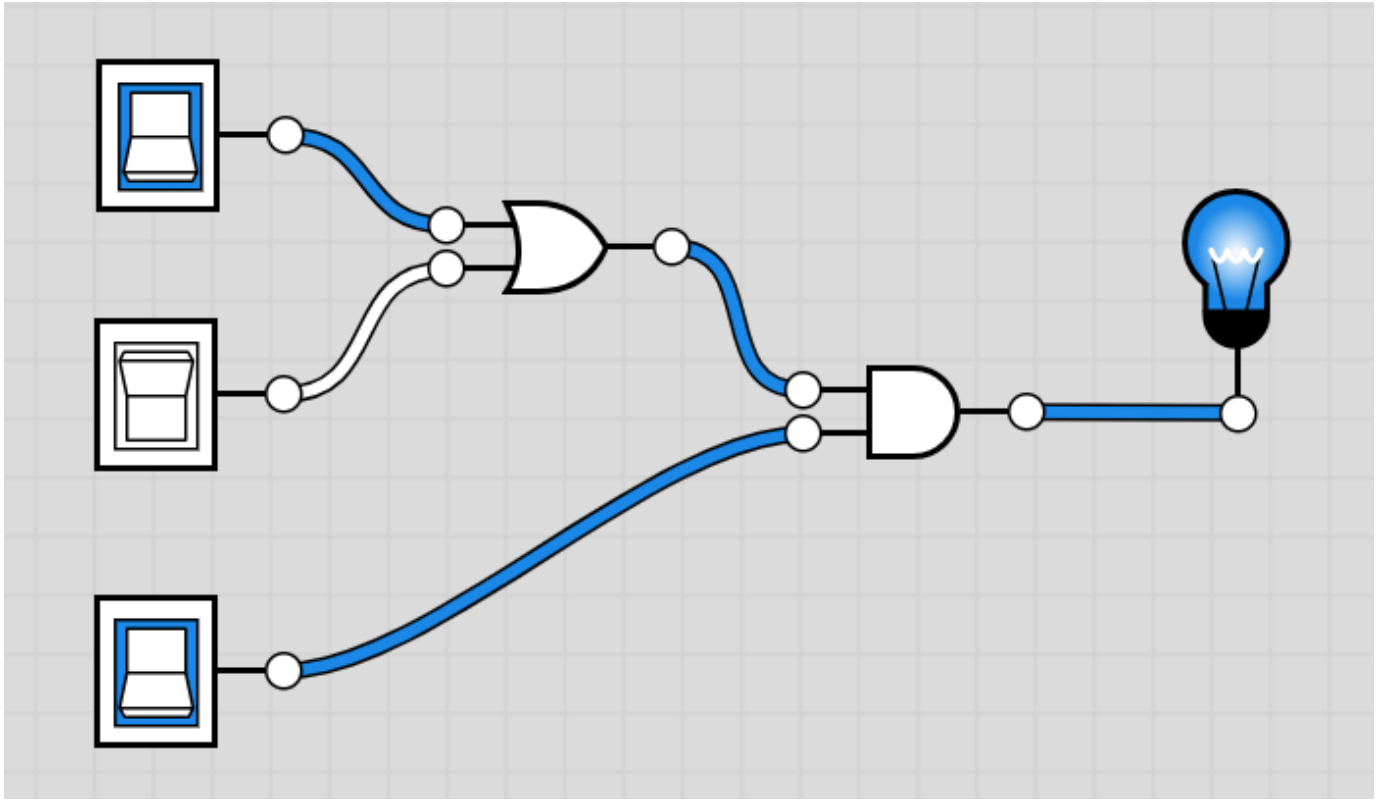
Example: $(A \vee \neg B)$



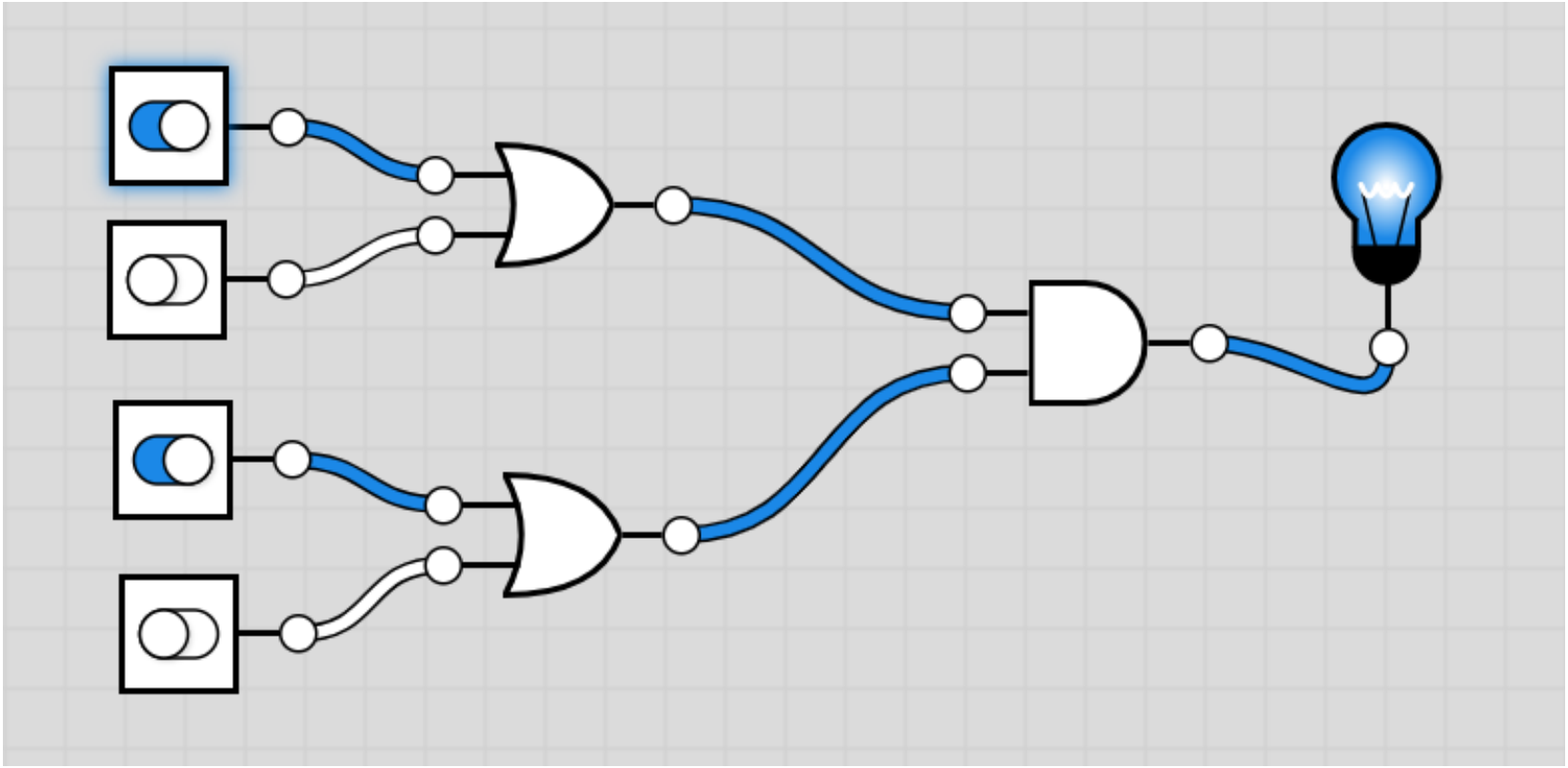
Example: $\neg(A \wedge B)$



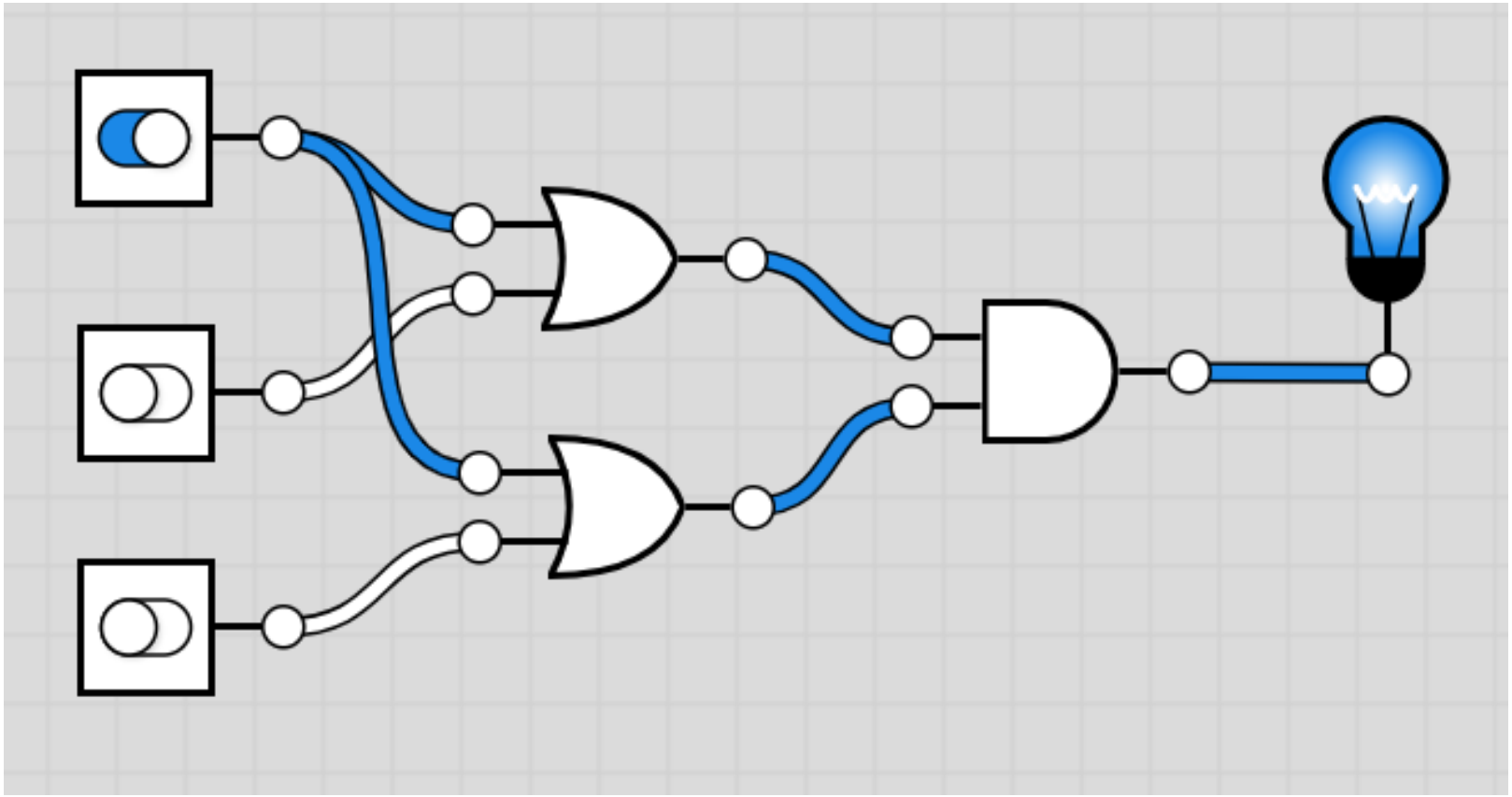
Example: $(A \vee B) \wedge C$



Example: $(A \vee B) \wedge (C \vee D)$



Example: $(A \vee B) \wedge (A \vee C)$

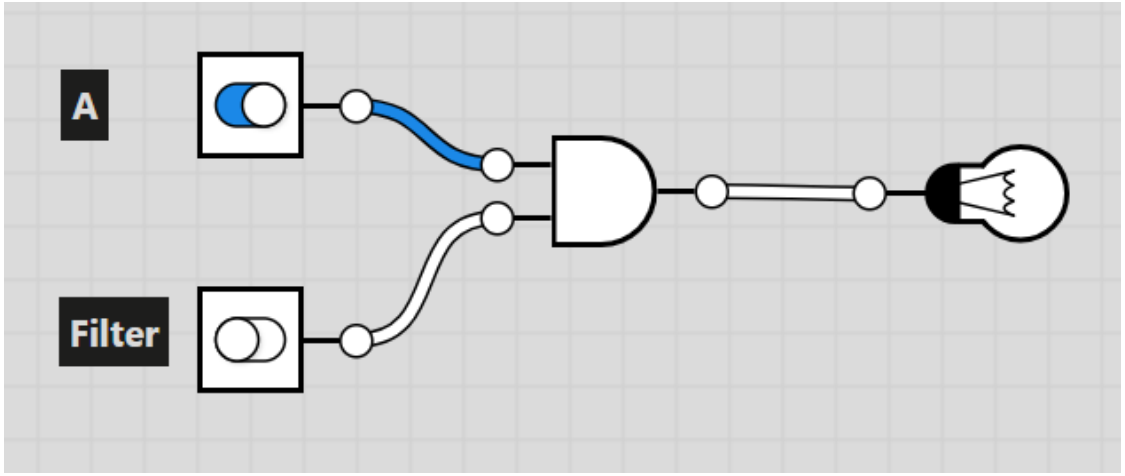


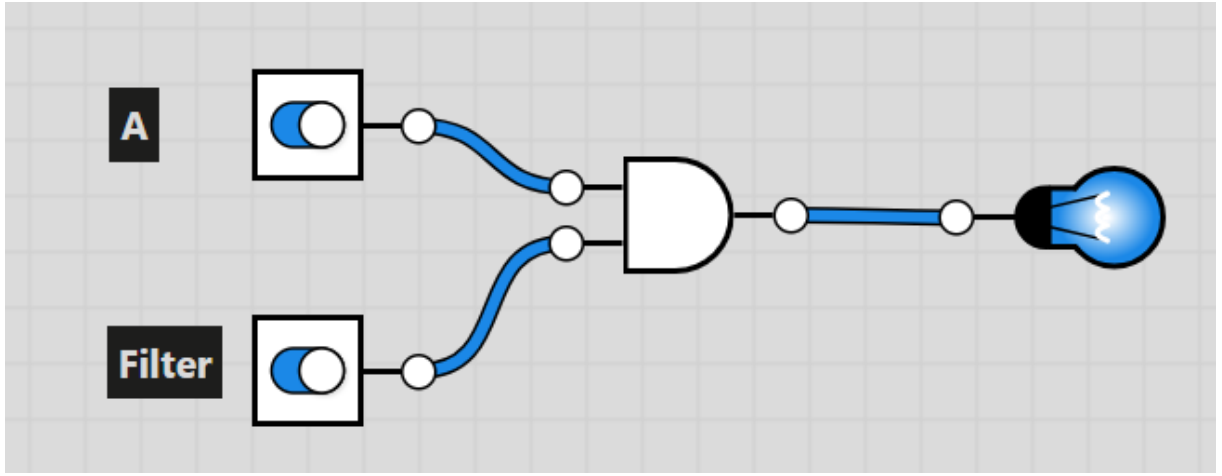
Exercise

- Navigate to https://logic.ly/demo/
  - Alternatively, do this using pen and paper.
- Construct logic circuits for the following:

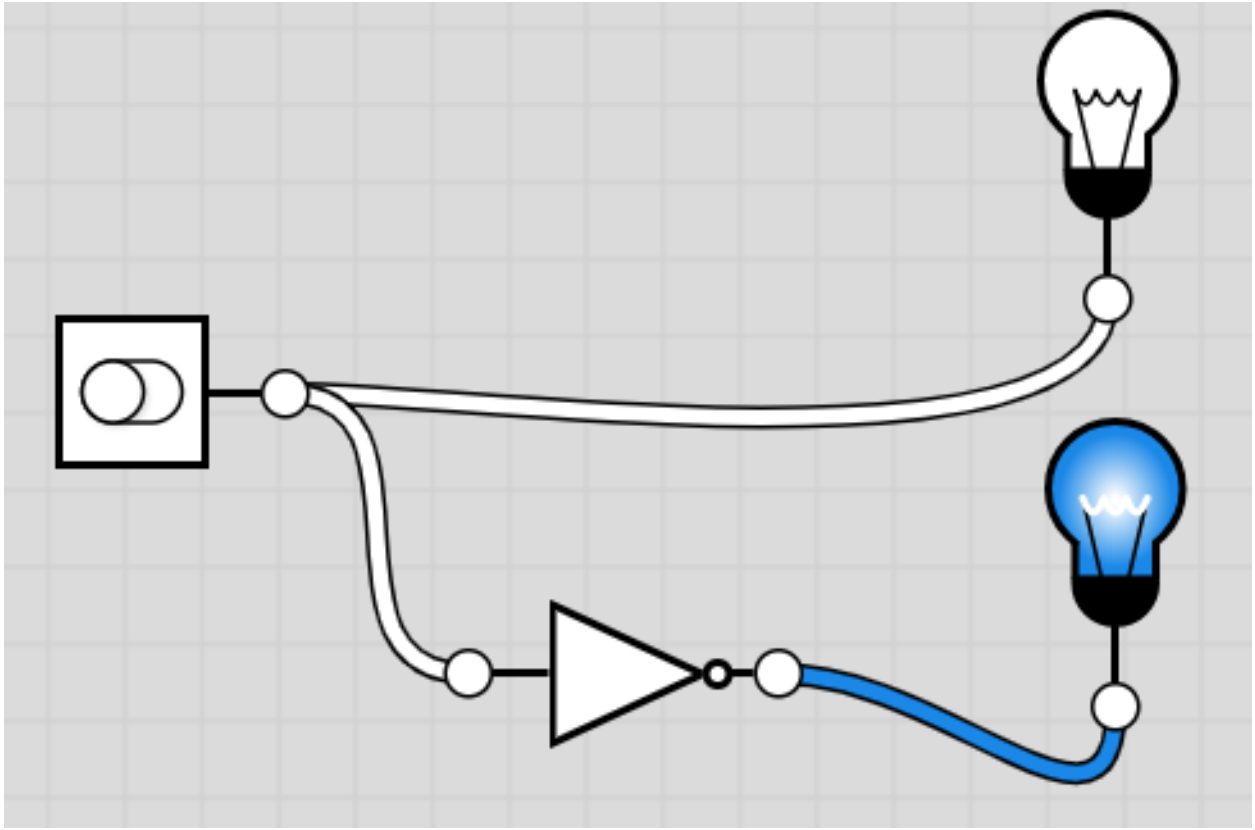
1. $(A \wedge B) \vee C$
2. $(\neg A \vee B) \wedge C$
3. $(\neg(A \wedge B)) \vee C$
4. $(A \wedge B) \vee (B \wedge C)$

Example: Using AND gate to filter an input





Example: Using NOT gate to create a selection input



Example: Using OR gate to output one of two inputs

